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Spectrogram Analysis for ECG Heartbeat Anomaly Detection Using a Novel Framework Incorporating Explainable AI with LIME and SHAP Techniques

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***Abstract* — Cardiovascular diseases are responsible for 17.9 million deaths annually, accounting for approximately 32% of global fatalities. This study introduces an automated framework for detecting cardiac anomalies across five heartbeat classes: normal, murmur, extrasystole, artifact, and extrahls from the Dangerous Heartbeat Disease dataset. The approach utilizes feature fusion of state-of-the-art deep learning models into a fine-tuned CNN-LSTM network, combining representations extracted from ResNet101, VGG16, and Inception V3 architectures. These features derived from Mel-frequency cepstral coefficients (MFCCs) and spectrogram analysis, capture critical acoustic patterns indicative of cardiac conditions. The core contribution lies in the integration of Explainable AI (XAI) techniques, SHAP and LIME to add a factor of interpretability to the system. This fine-tuned framework resulted in superior classification performance across key evaluation metrics, offering a transparent and efficient solution for cardiovascular diagnostics. The research lays a foundation for encouraging the trustworthy adoption of automated systems in clinical applications.**

***Keywords* — Heartbeat Anomalies, Electrocardiogram, Spectrogram Analysis, Explainable AI, SHAP, LIME.**

I. INTRODUCTION

The World Health Organization estimates that cardiovascular diseases account for approximately 17.9 million deaths annually. These figures highlight the need for early in mitigating the progression of such conditions to critical stages. Early diagnosis aids lifestyle modifications, medication or surgical procedures, thereby reducing the likelihood of severe complications like heart failure or stroke. Examples of common diagnostic tools used in this field include electrocardiograms (ECG) to monitor the heart's electrical activity, electronic stethoscopes for detecting murmurs indicative of blood flow or valve issues and pulse monitors for tracking irregularities in heart rate. These methods while effective require necessary expertise, are time-intensive and prone to human error. These challenges emphasize the need for automated systems capable of delivering more accurate diagnoses while minimizing human error. The proposed solution presents three advanced deep learning models, ResNet101, VGG16, and Inception V3, integrated and trained on spectrogram images derived from heart sound data.

The training process involves feeding spectrogram images to each model, enabling them to learn the acoustic features that distinguish normal from abnormal heartbeats. After individual model training, feature extraction is conducted to capture the learned representations from each network's layers. These features are then combined using feature fusion. Data preprocessing approaches are used in converting raw heart sound data collected through electronic stethoscopes into a structured format suitable for analysis. The resulting architecture, referred to as CNN-LSTM, benefits from these complementary attributes, resulting in a more complete understanding of the heart sound data. The combined CNN-LSTM model is fine-tuned post feature fusion and its weights are optimized for performance. One of the main challenges in deploying deep learning models in healthcare applications is the lack of interpretability. Since doctors require understanding into the decision-making process to ensure trust and accountability, the study incorporates Explainable AI (XAI) techniques to address this limitation. Two state-of-the-art XAI algorithms, SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) are integrated into the system. The purpose of this paper lies in the contribution of XAI in clinical contexts to reduce challenges of interpretability that is inherent in deep learning models.

The paper is organized to provide a clear and logical progression through the study. The next section reviews existing literature, highlighting gaps addressed by the proposed research. The methodology section introduces the proposed framework, detailing pre-processing, classification techniques, feature fusion and training parameters. The results section presents the key findings of the combined CNN-LSTM model and the XAI algorithms used in this study. The paper concludes by outlining potential future directions.

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II. LITERATURE SURVEY

Hassaan Malik et al. study abnormal heartbeat detection using RNN with LSTM, featuring MFCC extraction, noise reduction, and down-sampling. Their model achieves 99.71% accuracy on Pascal and 98.7% on PhysioNet, outperforming traditional models. Challenges include noise in datasets and reliance on labelled data[1]. S. Blok studied heartbeat detection using PPG technology with the Corsano 287 bracelet in 180 cardiac patients. PPG accuracy was 94.6% within 100 ms and 89.2% within 50 ms, with strong HR (R = 0.991) and RR (R = 0.891) correlations. Accuracy improved to 98.2% with a signal quality indicator, but further study is needed for movement scenarios [2]. Chethana et al. designed a Fiber Bragg Grating sensor system for cardiac monitoring across body positions. Advanced signal processing and custom software optimize the 1.20 pm strain-sensitive sensor for cardiac and respiratory parameters. The system supports remote health monitoring and wearable integration. Challenges include calibration and biocompatible coatings, but the study highlights its potential for personalized healthcare [3]. Alamr and Artoli's research on transformer-based anomaly detection in ECG signals combines transformers with GANs for feature extraction. Using two encoder and decoder blocks, the model achieves 98.2%-98.8% accuracy, outperforming traditional methods on the ECG 5000 testing set. Key achievements include enhanced anomaly detection and superior ROC curve performance [4]. Li and Boulanger's study detects structural anomalies in ECG signals using STFT spectrograms and handcrafted features. Their novel CNN architecture achieved 99.79% accuracy, 99.74% sensitivity, and a 0.15% false-alarm rate for 16 rhythm anomalies. The ResNet18-Gray model outperformed state-of-the-art methods, with a feature-merging layer boosting accuracy [5].

Tsimenidis, Lagkas, and Rantos explore deep learning for IoT-based ECG anomaly detection, achieving 98% accuracy in a cloud healthcare framework. Their method reduces signal length by 95%, improving processing speed and multiclass classification. Despite its success, the lack of validation on hospital data limits real-world applicability [6]. Karthikeyan, Murugappan, and Yaacob developed a FrFT-based algorithm for peak detection and feature extraction in ECG signals, achieving 99.8% accuracy on the MIT-BIH database. The method outperforms TERMA in peak detection and classifies CVDs using MLP and SVM with fewer features. However, accuracy drops to 38.2% in cross-database testing on SPH. Despite this, the study contributes an efficient algorithm for ECG signal analysis [7]. A study in Bioengineering 2022 developed an optimized 2D CNN-LSTM ensemble model for arrhythmia classification, achieving 98.7% accuracy for ARR, 99% for CHF, and 99% for NSR. Using wavelet transforms, 1D ECG data was converted to 2D scalograms for training, with K-fold validation ensuring robustness. The method avoids overfitting and effectively extracts sequential information but is computationally expensive. The study offers insights into optimizing ensemble models for high-accuracy arrhythmia classification [8]. Muhammad Zubair and Changwoo Yoon address anomaly detection in imbalanced ECG data using CNNs. Their method includes a temporal transition module and a cost-sensitive loss function to handle data imbalance. The model achieved 99.81% accuracy for intra-patient and 96.36% for inter-patient classification on the MIT-BIH dataset. Strengths include handling imbalanced data without overfitting, though inter-patient classification faces challenges due to morphological variability [9]. June Young Park et al. develop EB-GAME, a novel framework for ECG anomaly detection using only normal signal labels for training. The framework achieves 0.97 accuracy, 0.97 AUROC, and 0.95 F1-score, outperforming models like AnoGAN and BeatGAN. Its main contribution is the effective anomaly detection using normal data alone [10].

Collectively, these studies highlight the significant advancements in ECG-based anomaly detection and arrhythmia classification using deep learning, machine learning, and anomaly detection frameworks. These studies often face challenges with reliance on large annotated datasets and issues with data imbalance. Many of these models lack explainability or have challenges with the performance of rare anomalies. The proposed framework aims to address these gaps by integrating a hybrid approach that offers both high performance and practical applicability for real-time heart anomaly detection.

III. PROPOSED METHODOLOGY

A. Dataset Preparation

The dataset utilized in this study is the Dangerous Heartbeat Dataset (DHD), an open-source collection sourced from Kaggle. This dataset integrates recordings from two primary sources: the general public, captured via the iStethoscope Pro iPhone application, and clinical trials conducted in hospitals using the DigiScope digital stethoscope. It comprises 661 labelled audio files representing a diverse range of heart conditions. It is important to note that the DHD dataset has inherent limitations due to its mixed sourcing. Recordings collected from public sources using mobile applications and clinical trials with digital stethoscopes can differ significantly in quality and recording conditions. These variations may introduce sample selection bias and inconsistencies in data collection processes. There exist five primary categories of heart sounds in this dataset: normal, murmur, extra systolic, artifact, and extrahls.

The study employs data augmentation techniques to ensure balanced representation across all classes. Heart sound recordings vary in duration. The recordings are flattened. Flattening involves adjusting the sample rate to maintain a consistent temporal resolution. One-hot encoding technique was then applied to categorize each class explicitly. The technique converts categorical labels into binary vectors. The next step involves noise reduction, a measure to boost signal clarity. Initially, band-pass filtering was employed to isolate the frequency ranges relevant to heart sounds, eliminating extraneous background noise. This step was followed by wavelet decomposition, used to separate noise from the primary signal by breaking it into distinct components. Once the data is cleaned and standardized, it is transformed into a spectrogram - a visual representation of sound frequencies over time. This conversion produces a heat map that captures the frequency, amplitude, and temporal evolution of the heart sounds in a two-dimensional format. The pipeline ensures that the heart sound data is prepared for efficient analysis in the next stage.

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B. *Model Architecture*

The study combines three state-of-the-art deep learning models - ResNet101, VGG16, and Inception V3 to analyze and classify heartbeat sound signals, each chosen for its unique strengths in feature extraction. ResNet101 proves to be adept at identifying subtle patterns in heart sounds. VGG16 with its simple yet powerful architecture to capture basic signal characteristics. Inception V3 to detect both fine and coarse patterns in the heart sound data. Following the training of individual models, feature extraction is performed to capture the learned representations from each network's layers. Mel Frequency Cepstral Coefficients (MFCCs) are used for feature extraction. Features extracted by the fine-tuned CNN components were subsequently fed into multiple LSTM layer network arranged in a stacked configuration, forming a CNN-LSTM hybrid architecture. The combined architecture benefits from these complementary attributes, resulting in a more complete understanding of the heart sound data. The core architecture as depicted in Figure 1, integrates convolutional and pooling layers, ensuring that the sequential nature of heart sound data is effectively analyzed. The architecture's primary strength lies in the two stacked LSTM layers which are adept at capturing long-range dependencies and temporal dynamics inherent in sequential data. The final output layer employs a softmax activation function for multiclass classification across the five categories.

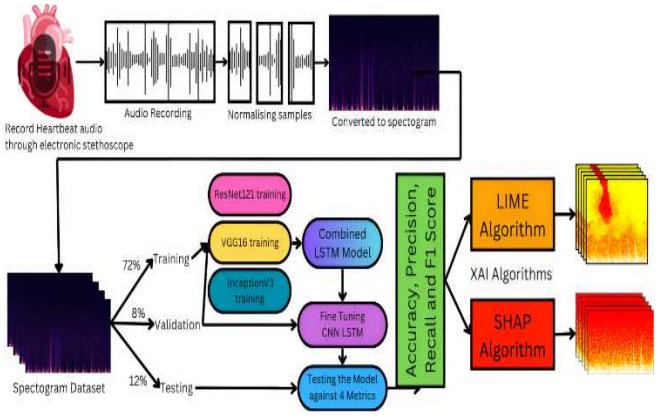


Fig. 1. Proposed Model Architecture for Anomaly Classification

The model was optimized during the training process using the Adam optimizer with a learning rate of 0.0001 and the categorical cross-entropy loss function. Hyperparameter tuning was also performed to optimizes key variables ensuring optimal performance on the validation set. The framework used early stopping with a patience threshold of 15 epochs to prevent overfitting along with a model checkpointing mechanism to ensure optimal model selection during the training process. The CNN-LSTM architecture was fine tuned to model the temporal dependencies in heartbeat sound signals. The fine-tuning process plays a critical role in improving the model's capacity to adapt to subtle distinctions in the dataset without compromising the generalized features learned in earlier layers.

C. *Model Training and Evaluation*

The model's performance is evaluated against five baseline architectures that were tested. The evaluation metrics include accuracy, precision, recall, and F1 score. A novel aspect of this study is the incorporation the Explainable AI techniques. XAI consists of various algorithms designed to render machine learning models comprehensible by clarifying the reasoning behind their predictions. In this study, two of the top-performing XAI techniques have been applied: SHapley Additive Explanations (SHAP) which assigns importance scores to each feature based on its contribution to the prediction. LIME (Local Interpretable Model-Agnostic Explanations), a technique designed to generate explanations for individual predictions, provides a localized understanding of the model's decision-making process. Both SHAP and LIME serve complementary purposes in model interpretability. Together, these methods allow for a multi-faceted understanding of model behavior.

IV. RESULTS AND DISCUSSION

The findings of this study affirms the efficiency of the fine-tuned CNN-LSTM architecture in detecting and classifying anomalies within heartbeat sounds. The data was divided into 72% for training, 8% for validation, and 20% for testing, ensuring a balanced approach to model learning and evaluation. The audio data was transformed into Mel Frequency Cepstral Coefficients (MFCCs). The below analysis shows the unique spectral and temporal characteristics of two heartbeat classes. Visualizations for the normal and the extrasystole classes including spectrum plots, waveforms, spectrograms and MFCCs provides detailed information into the acoustic properties of different heartbeat sounds.

The spectrum plots for normal heartbeat sounds in Figure 2 reveals dominant energy in the lower frequencies. The waveform displays a regular, periodic pattern with consistent amplitude characteristic of the "lub-dub" sound. Spectrogram images illustrate the uniformity of energy distribution over time and frequency. The extrasystole class demonstrated in Figure 3 shows periodic patterns with sporadic anomalies in the waveform, indicative of premature or extra beats. The spectrogram reveals dominant energy in the low-frequency range, with secondary peaks corresponding to irregular energy contributions from the extra beats.

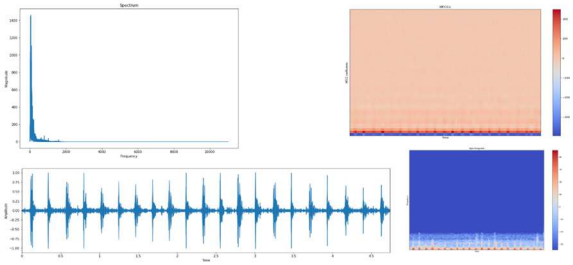


Fig. 2. Spectrum Plots and Waveforms for the normal class

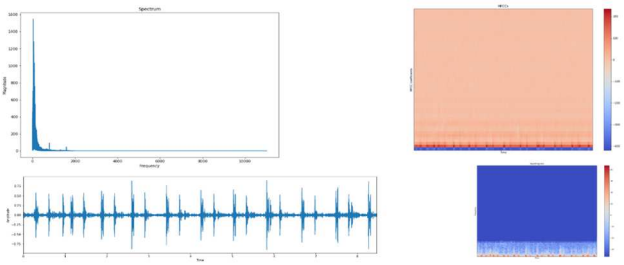


Fig. 3. Spectrum Plots and Waveforms for the extrasystole class

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The training and loss curves post-fine-tuning i.e. after the last 50 layers are kept trainable and tested as shown in Figure 4, the training curves smoothed considerably, and validation loss demonstrated a consistent downward trend, showcasing improved generalization. This improvement emphasizes the importance of fine-tuning in optimizing the model for the specific nuances of the heartbeat audio.

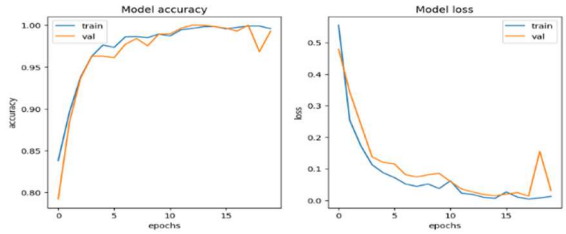


Fig 4. Accuracy and Loss curves after fine-tuning the model

The results indicate that the proposed model outperformed other tested architectures - SimpleRNN, Temporal Convolutional Network (TCN), Gated Recurrent Unit (GRU), Multi-Layer Perceptron (MLP), and a regular CNN-LSTM, across all key metrics. The model achieved an accuracy of 92%, precision of 92%, recall of 93%, and an F1 score of 92%. These metrics highlight superior performance which is crucial in healthcare environments. Table 1 along with the radar plot in Figure 5 (a) compares the fine-tuned CNN-LSTM's metrics against other models, showcasing its superiority across metrics. The fine-tuned CNN-LSTM demonstrates near-perfect values, reinforcing its suitability for heart sound classification. The ROC curve in Figure 5(b) evaluates the fine-tuned CNN-LSTM's classification performance for critical heart sound classes. The model achieves an Area Under the Curve (AUC) of 1.00 for the normal class, signifying perfect classification. These high AUC scores highlight the model's ability to maintain a balance between sensitivity and specificity, making it well-suited for cardiac health assessment.

Model	Accuracy	Precision	Recall	F1 Score
Fine-Tuned CNN LSTM	0.92	0.92	0.93	0.92
Simple RNN	0.88	0.87	0.86	0.86
TCN	0.89	0.88	0.87	0.87
CNN-LSTM	0.90	0.89	0.89	0.89
MLP	0.85	0.84	0.83	0.83

Table 1. Comparative Analysis of the models

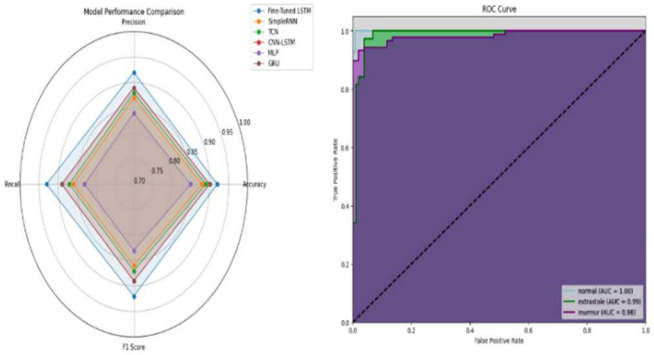


Fig 5. (a) Radial Plot of the Evaluation metrices (b) ROC curve

The incorporation of SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) into the study introduces a factor of interpretability into the model for deeper understanding of its decision-making process. SHAP assigns individual feature importance scores, identifying critical attributes in the spectrogram such as variations in frequency, pitch, and amplitude, which are strongly indicative of anomalies. LIME provides insights into individual predictions by approximating the model's decision boundary using an interpretable local surrogate model. Both the Explainable AI techniques play vital roles in this study by offering complementary perspectives.

Figure 6 showcases an original spectrogram of the normal class on the left. The rhythmic pattern characteristic of a healthy heartbeat is evident, with consistent intervals and amplitude representing the regularity of the cardiac cycle. This class serves as a baseline for identifying anomalies and understanding non-pathological heart rhythms. The absence of irregular spikes or deviations highlight the model's precision in distinguishing normal conditions.

Adjacent to the original image, the LIME explanation in Figure 6 reveals localized areas of importance in the spectrogram. Darker highlighted segments correspond to the "lub-dub" frequency range, illustrating the ability of the model to focus on the most critical components of normal heart sounds for classification. In contrast, the SHAP explanation presented in Figure 7 identifies the features that most influence the fine-tuned CNN-LSTM. The presence of regular and periodic spikes offers a global view of the factors driving predictions.

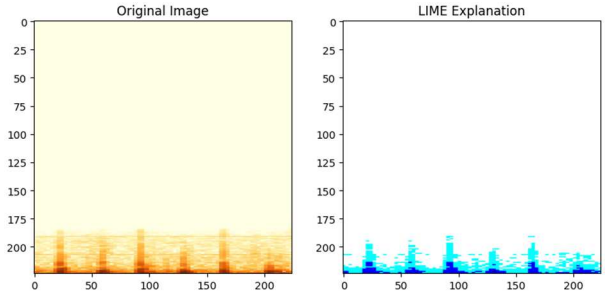


Fig 6. Original Spectrogram and LIME explanation of the normal class

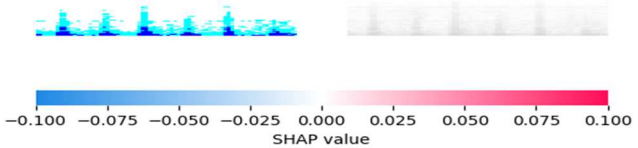


Fig 7. SHAP Heatmaps for the normal class

The class that can be examined thereafter is an extrasystole. It may not be a sign of disease. It can happen normally in an adult and can be very common in children. In Figure 8, the spectrogram for the extrasystole class highlights irregularities in the usual "lub-dub" pattern. Extrasystoles are characterized by sudden deviations in both amplitude and frequency. This anomaly appears as irregular spikes and

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additional beats, a visual differentiation crucial for the model's detection of the extrasystole class.

The LIME explanation marks lighter regions in the spectrogram, where deviations from the normal heart rhythm are most pronounced. These highlighted areas correspond to the premature contractions and they play a key role in classifying the sample as an extrasystole. The SHAP heatmap in Figure 9 illustrates the overall influence of abnormal patterns like the out of rhythm heartbeats and occasional skipped beats which are observed as non-periodic flows and sudden spikes within the spectrogram on the model's predictions emphasizing their contrast with the surrounding normal intervals.

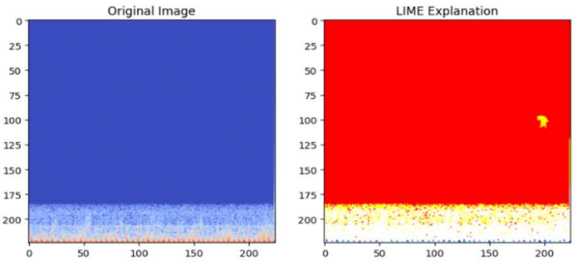


Fig 8. Original Spectrogram and LIME explanation of the extrasystole class

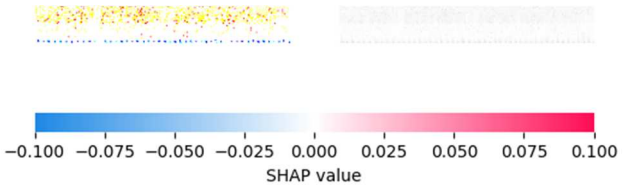


Fig 9. SHAP Heatmaps for the extrasystole class

The synergy between LIME's localized insights and SHAP's broader perspective, both incorporated into the proposed framework improved the model's applicability by ensuring reliable decision-making in clinical settings.

V. CONCLUSION AND FUTURE WORK

The study presents a comprehensive approach to heartbeat anomaly detection, combining state-of-the-art machine learning models and techniques with Explainable Artificial Intelligence (XAI) methods to ensure robust, accurate, and interpretable outcomes. The training of the proposed model architecture demonstrates its efficacy through superior performance metrics, including high classification accuracy to all the five classes, namely normal, murmur, extrahls, extrasystoles and artifact. The model's architecture is designed to balance complexity and efficiency, enabling it to process intricate patterns in heartbeat data while maintaining computational feasibility.

The integration of XAI techniques, SHAP and LIME provides critical insights into the decision-making process of the model. The dual application of SHAP and LIME proves to be valuable for cardiac anomaly detection, as these methods provided complementary perspectives. The proposed methodology bridges the gap between automated diagnostics and clinician trust. This work represents a significant step toward the adoption of reliable, interpretable diagnostic tools in the medical domain, aiming to enhance patient care and the capabilities of modern healthcare. Despite the promising results in controlled experimental environments, the real-world applicability of the framework may be constrained by factors such as device variability, heterogeneous recording conditions and sample selection bias.

Future work should prioritize extensive external validation and the incorporation of more controlled, diverse datasets to mitigate these biases and ensure robust generalization across various clinical settings. It can explore further optimization of the model for real-time applications and its generalization to broader physiological datasets and validating its performance in diverse clinical settings.

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